

## Employment

## Product Engineer Intern

## Isomer AI

**May 2025 – May 2026**

- Ship 10+ production AI features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflow orchestration, MCP tooling, real-time event pipelines (Pusher), Auth0 RBAC; 923 commits, 7 deployments.
- Own PostgreSQL, Redis, Docker, Sentry; build document processing pipelines and resolve production incidents.

## Graduate Research Assistant

**NC State University**

**April 2025 – Present**

- Deploy cloud genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets.
- Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

## Software Developer Intern

## Mphasis (HP Company)

**Jun 2024 – Aug 2024**

- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%.
- Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.

## AI/ML Intern

## Capgemini

**Jun 2023 – Aug 2023**

- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%.
- Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.

## Education

**Raleigh, NC**

**North Carolina State University**

**Aug 2024 – May 2026**

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.

**Mumbai, India**

**Mumbai University**

**Aug 2020 – May 2024**

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.

## Projects

## LLM Eval Harness

## Independent Project

**2025**

- Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks.
- Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap CIs and McNemar tests across 6 metrics.

# KubeServe

## Kubernetes LLM Inference Platform

2026

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Instrumented Otel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.

## VectorLift

## Semantic Search & Ranking Engine

2026

- Built production-grade semantic search on MS MARCO: BM25 (Elasticsearch), dense bi-encoder (sentence-transformers, in-batch negatives), cross-encoder re-ranker (MiniLM); 6 pipeline configs, NDCG@10/MRR@10/MAP.
- Kafka-driven corpus indexing into Elasticsearch and Qdrant; FastAPI search API; Streamlit experiment dashboard.

## AlgoPulse

## Agentic AI Platform

2025

- Replaced API-polling bottleneck with Supabase Edge Functions streaming AI reasoning loops to React Native clients in real time, eliminating state loss under load; Temporal.io durable workflows; 2,000 req/sec.
- Orchestrated Temporal.io workflows and Kafka queues for fault-tolerant state sync; zero message loss under sustained load.

## Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- **AI & Agents:** Claude/GPT-4o/Gemini APIs, LangChain, MCP, RAG, sentence-transformers, Hugging Face
- **Evaluation & Search:** LLM benchmarking, NDCG/MRR metrics, Elasticsearch, Qdrant, PostgreSQL, Redis
- **Infra & Testing:** Docker, Kubernetes, Kafka, Temporal.io, FastAPI, GitHub Actions, Prometheus

### Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at [kulkarniaditya.com/blog](https://kulkarniaditya.com/blog)